K NEAREST NEIGHBOUR

Wine Quality Classification

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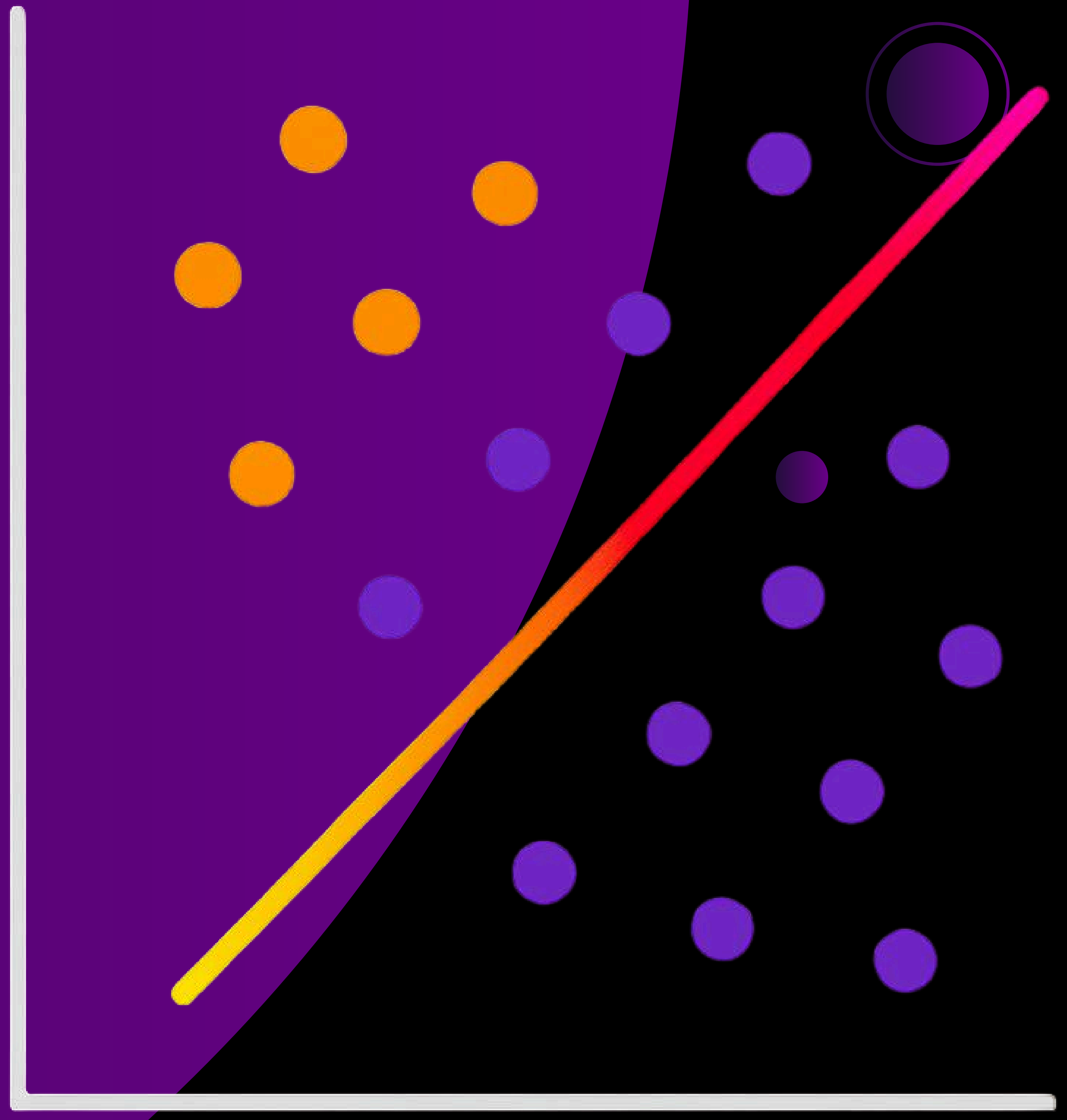
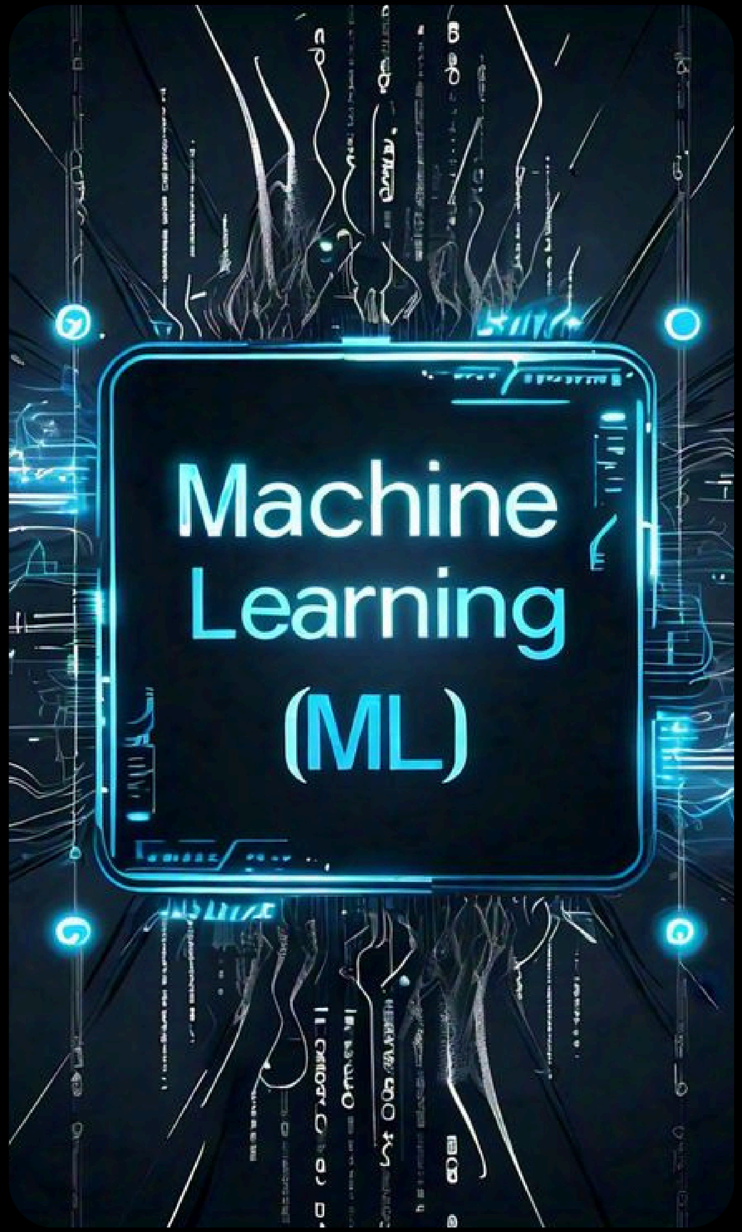


MACHINE LEARNING



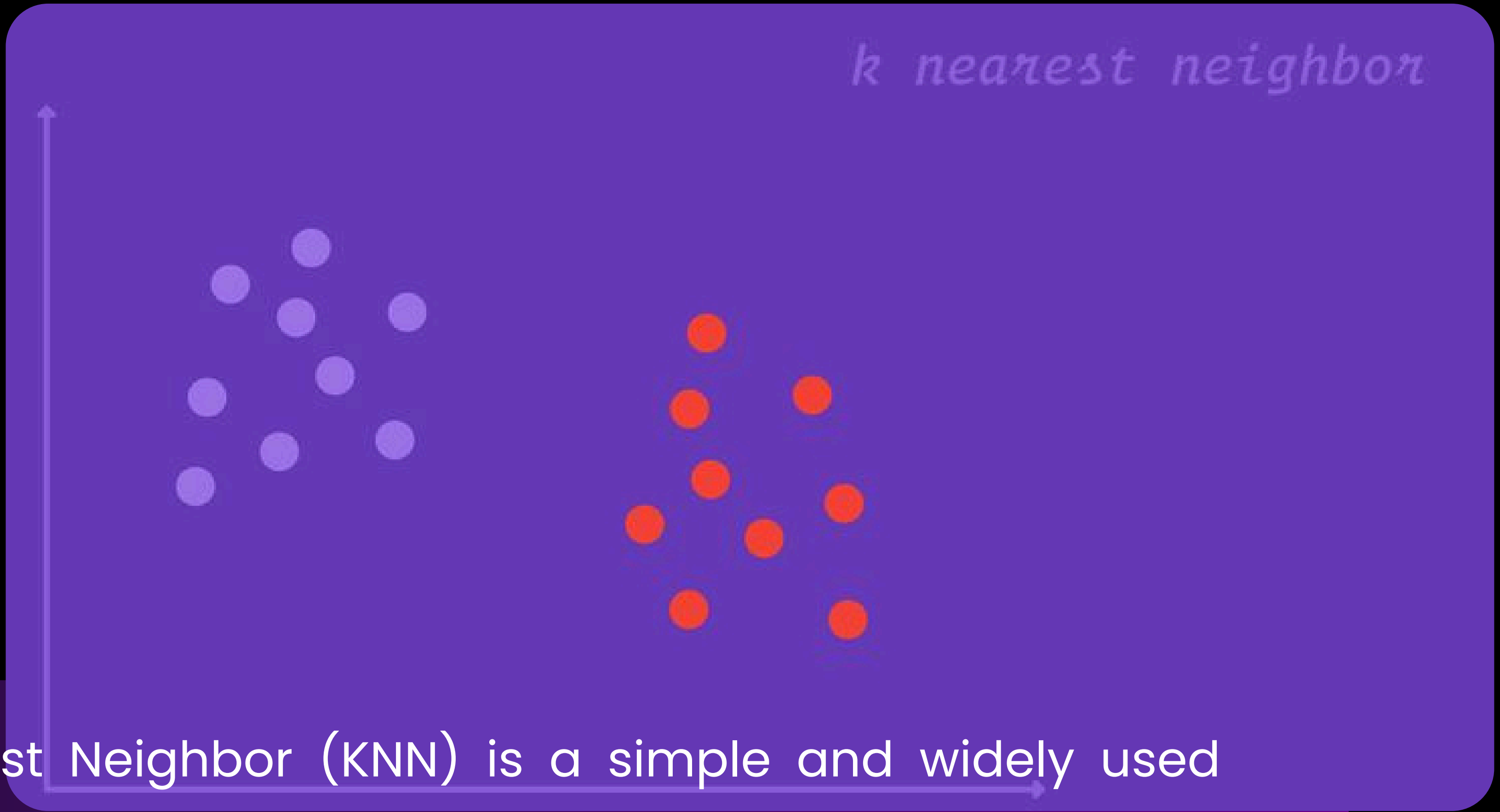
Machine Learning is a technique that allows computers to learn from data and make decisions without explicit programming. It works by identifying patterns in data and using them to make predictions.

This learning happens through the following steps:

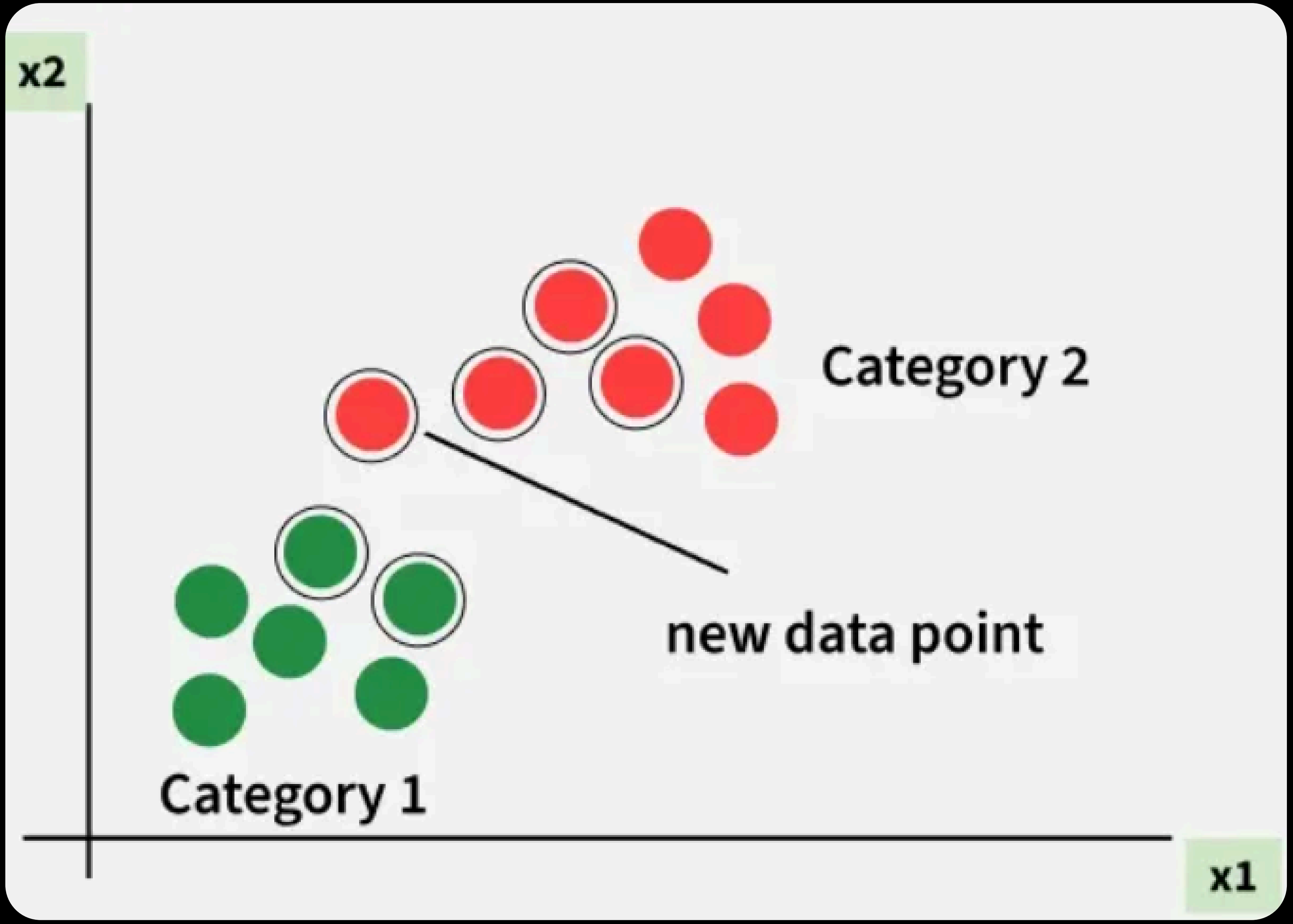
Data Input
Algorithms
Model Training
Feedback Loop
Experience and Iteration
Evaluation and Generalization

INTRODUCTION – KNN

1. Classifies data based on similarity with nearby data points
2. Uses distance metrics like to find nearest neighbors
3. Since KNN makes no assumptions about the underlying data distribution, it makes it a non-parametric and instance-based learning method.



K-Nearest Neighbor (KNN) is a simple and widely used machine learning technique for classification and regression tasks. It works by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbors.



KNN

KNN assigns the category based on the majority of nearby points. The image shows how KNN predicts the category of a new data point based on its closest neighbours.

The green points represent Category 1 and the red points represent Category 2.

The new data point checks its closest neighbors (circled points).

Since the majority of its closest neighbors are red points (Category 2) KNN predicts the new data point belongs to Category 2.

PROBLEM STATEMENT.

The wine industry is highly competitive, where consistent product quality is essential for:

- Brand reputation
- Customer satisfaction
- Profitability

Traditional quality assessment depends on human experts (sommeliers), which is:

- Time-consuming
- Subjective
- Costly

This project aims to develop a predictive model using K-Nearest Neighbors (KNN) to:

- Classify wine quality as Good or Bad
- Use physicochemical properties of wine

Business Objective:

To support wineries in improving quality control processes and ensuring consistent product standards using data-driven insights.

DATASET OVERVIEW

SOURCE:

- Kaggle
- Publicly available and widely used for Machine Learning tasks

DOMAIN CONTEXT:

- Industry: Food & Beverage (Wine Production)
- Type: Classification Problem
- Goal: Predict wine quality using Chemical attributes

DATASET SIZE:

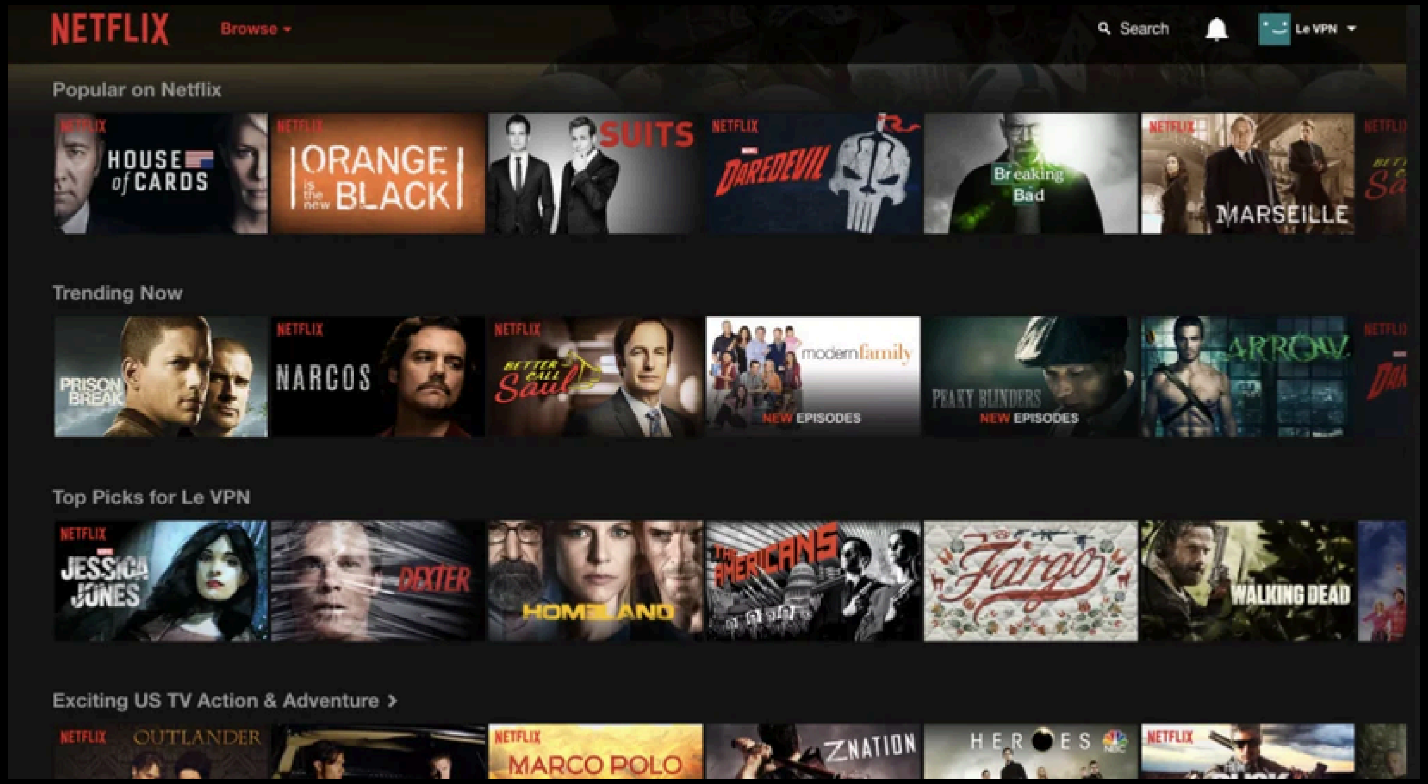
- Contains 1500 rows(observation)
- 11 input features + 1 target variable (Quality)

VARIABLES (features):

- Fixed acidity, Volatile acidity
- Citric acid, Residual sugar
- Chlorides, Alcohol
- Free Sulfur dioxide, Total Sulfur dioxide
- pH, Sulphates
- density

REAL-LIFE APPLICATION

KNN in Netflix / YouTube



- Users represented as data points
- Based on watch history and preferences
- Finds similar users (nearest neighbors)
- Recommends content watched by similar users
- Enables personalized recommendations

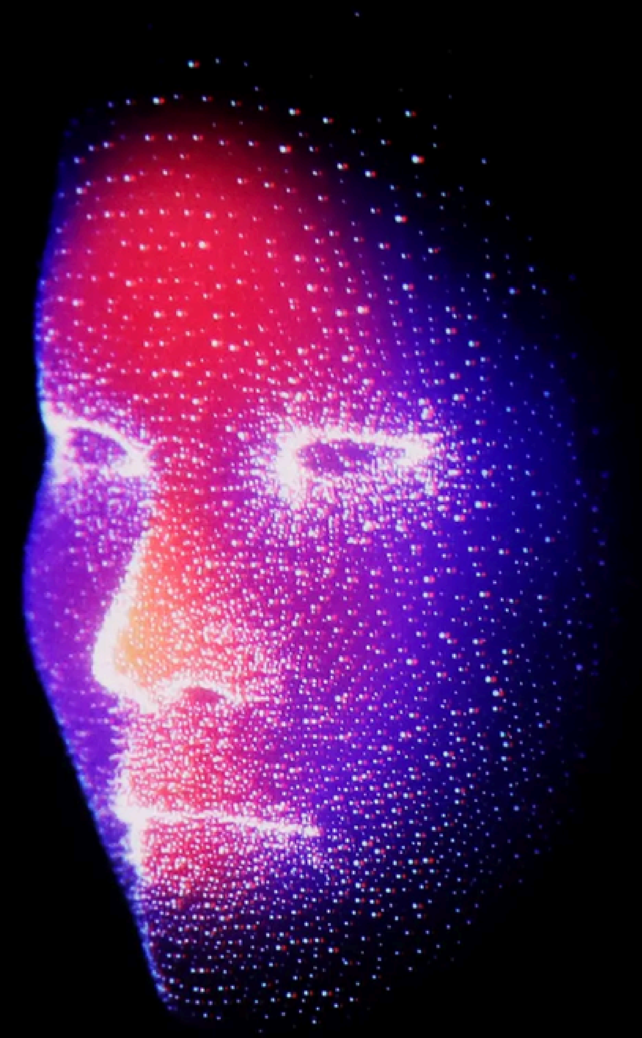
REAL–LIFE APPLICATION

KNN in E-commerce (Amazon)

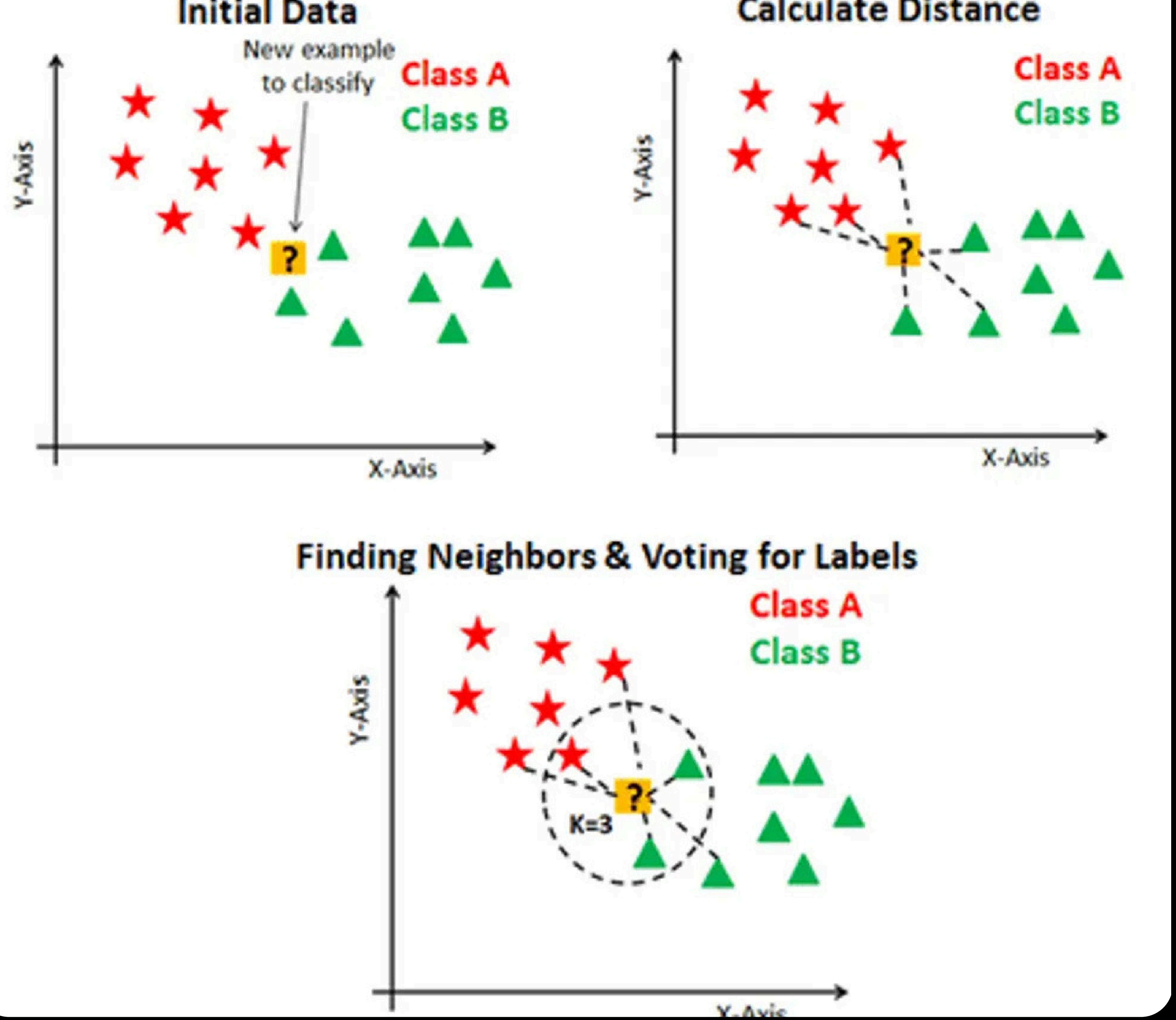
- Customers represented as data points
- Based on purchase & browsing history
- Identifies similar customers
- Recommends products bought by others
- Increases sales and user engagement

REAL-LIFE APPLICATION

KNN in Face Recognition



- Converts face into numerical features
- Stores known faces as data points
- Compares new face with stored data
- Finds closest match using KNN
- Used in phone unlock and security systems



WORKFLOW OF KNN

1. **Selecting the optimal value of K (using Cross-Validation)**

- **K** = Number of nearest neighbors used for prediction
- **Large K** → Simpler model, may miss patterns
- **Small K** → Sensitive to noise and outliers
- **Noisy data** → Use larger K for stable predictions
- K controls the balance between overfitting and underfitting

2. **Calculating distance**

- For a new data point, calculate the distance from all existing data points. Comman method: **Euclidean Distance**

3. **Finding Nearest Neighbors**

- Sort all distances in ascending order
- Select the K closest data points

4. **Majority Voting (Classification)**

- Count the classes of the K neighbors
- Assign the class with the highest frequency
- This process is called Majority Voting

Cross Validation:

A technique used to evaluate a model by splitting the data into multiple parts and testing it on different subsets. It ensures the model performs well on unseen data and is not dependent on a single train-test split.

WINE QUALITY CLASSIFICATION USING PYTHON

Basic Library used in code:

Pandas

NumPy

seaborn

matplotlib

Scikit-learn (sklearn)

GridSearchCV

StandardScaler

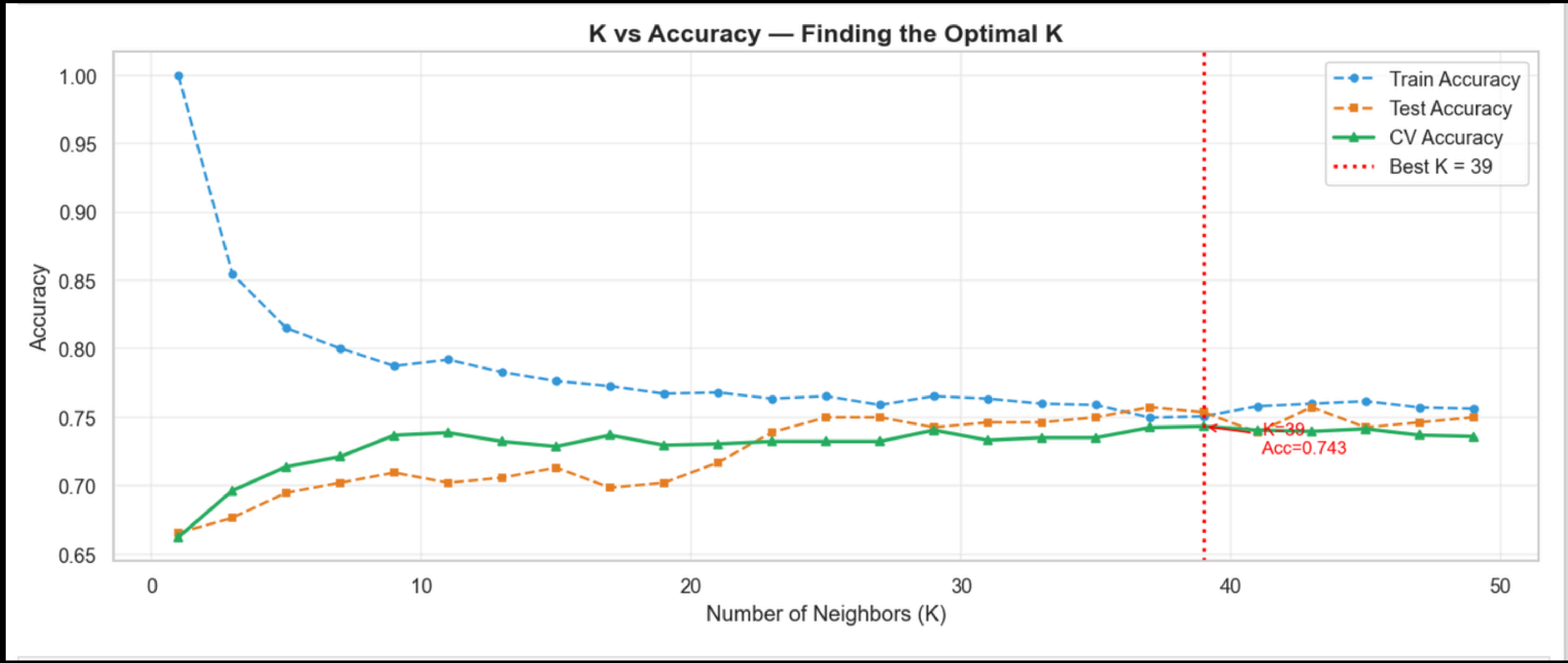
cross_val_score

LabelEncoder

INTERPRETATION

Target Distribution

Finding the Best K value



INTERPRETATION

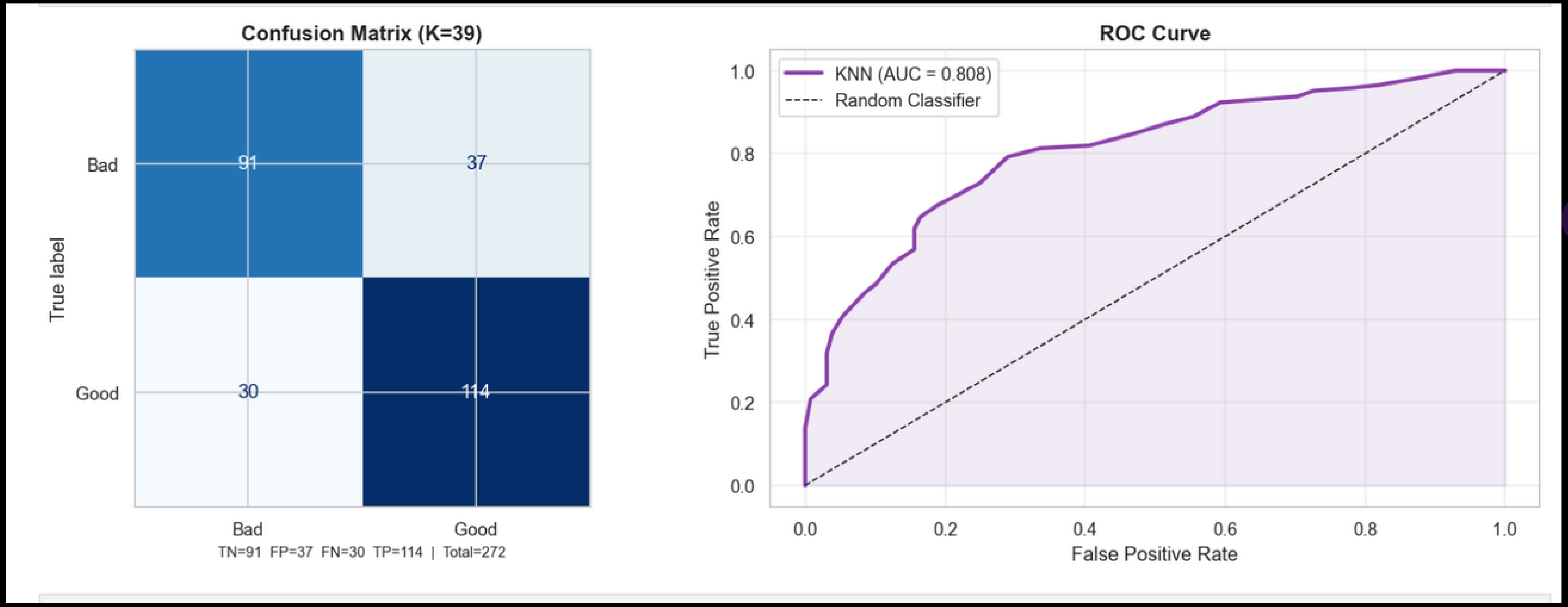
Classsification Report

```
# Detailed classification report
print("\nCLASSIFICATION REPORT")
print("=" * 50)
print(classification_report(y_test, y_pred, target_names=['Bad', 'Good']))
```

```
CLASSIFICATION REPORT
==================================================
              precision    recall  f1-score   support

         Bad       0.75      0.71      0.73       128
        Good       0.75      0.79      0.77       144

    accuracy                           0.75       272
   macro avg       0.75      0.75      0.75       272
weighted avg       0.75      0.75      0.75       272
```



Confusion Matrix

INTERPRETATION

MODEL PERFORMANCE SUMMARY

Metric	Score	Interpretation
Accuracy	0.7537	Overall correct predictions
Precision (Good)	0.7550	Of predicted "good", how many are actually good
Recall (Good)	0.7917	Of actual "good", how many were caught
F1-Score (Good)	0.7729	Harmonic mean of precision and recall
ROC-AUC	0.8082	Discriminative ability across all thresholds
CV Mean Accuracy	0.7226	Robust estimate across 5 folds

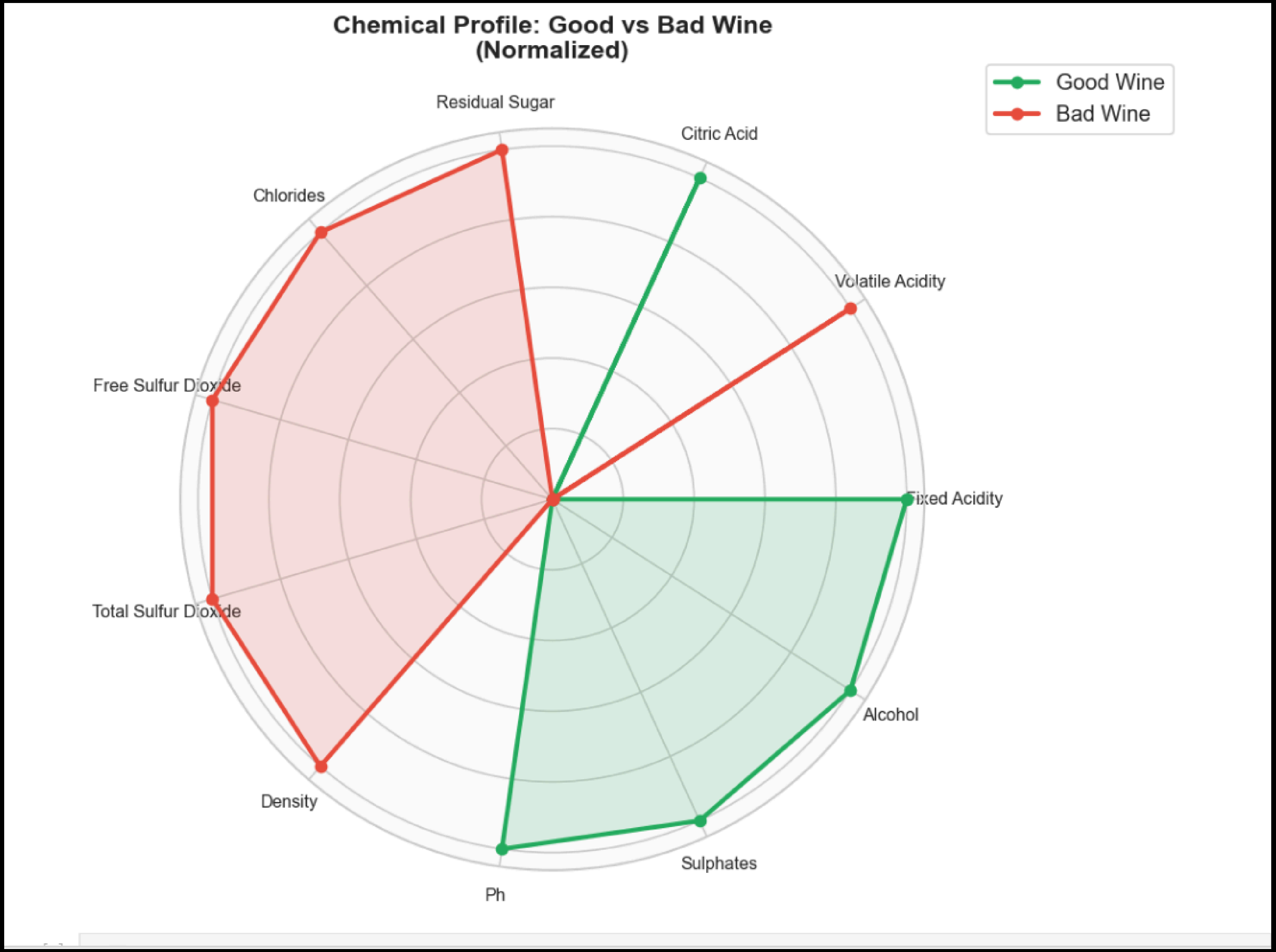
Model Performance

MEAN CHEMICAL PROPERTIES BY QUALITY CLASS

quality	bad	good	Difference	Driver
alcohol	9.919	10.883	0.964	↑ Good
fixed acidity	8.118	8.432	0.314	↑ Good
sulphates	0.609	0.687	0.078	↑ Good
citric acid	0.237	0.303	0.066	↑ Good
pH	3.309	3.309	0.000	↓ Bad
density	0.997	0.996	-0.001	↓ Bad
residual sugar	2.327	2.321	-0.006	↓ Bad
chlorides	0.085	0.078	-0.007	↓ Bad
volatile acidity	0.590	0.473	-0.117	↓ Bad
free sulfur dioxide	16.445	15.064	-1.381	↓ Bad
total sulfur dioxide	54.298	38.789	-15.509	↓ Bad

Identification of best chemical properties

INTERPRETATION



BUSINESS INSIGHTS

Automated Quality Control

→ KNN model (~75% accuracy) can be used to predict wine quality before sale, reducing manual testing.

Key Quality Drivers Identified

→ Higher alcohol, sulphates, and balanced acidity improve quality, while high volatile acidity and sulfur reduce it.

Production Optimization

→ Insights help in adjusting chemical composition during fermentation to consistently produce better-quality wine.

Cost & Waste Reduction

→ Early detection of low-quality wine helps avoid packaging, storage, and market losses.

Better Pricing & Market Strategy

→ Enables segmentation into premium and low-tier products, improving revenue and brand positioning.

FUTURE ACTIONS

Key Strategies for Model Improvement

- Feature Scaling (Crucial Step)
- Dimensionality Reduction
- Adding multi-class classification
- Integration of multiple wine dataset

THANK YOU